

# Final Report: A Brain-Computer Interface Control System Design Based on Deep Learning

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## Abstract

Motor imagery (MI) Brain-Computer Interfaces (BCIs) remain difficult to deploy for real-world robotic control because existing approaches treat EEG decoding as a static, open-loop classification problem and depend on high-density electrode arrays. This project presents a closed-loop BCI control pipeline that addresses these limitations through three contributions: (1) a hybrid CNN–Transformer classifier (EEGTransformer) that captures both the spatial specificity and long-range temporal structure of MI signals, (2) a Deep Q-Network (DQN) reinforcement learning agent that compensates for transient classification errors by integrating evidence over time, and (3) a systematic channel reduction study that maps the 64-channel laboratory setup down to an 8-channel configuration compatible with consumer-grade OpenBCI hardware.

Evaluated across BCI Competition IV-2a (22-channel, 4-class), IV-2b (3-channel, 2-class), and PhysioNet EEGMMIDB (64-channel, 109 subjects), the EEGTransformer achieved mean accuracies of 73.80%, 82.87%, and 88.78% (with subject-specific fine-tuning), respectively. The channel reduction study identified C3 as the most critical electrode and showed that an 8-channel subset retains 72.54% accuracy after fine-tuning. A Transformer-based DQN agent achieved a 100% target reach rate in simulation, and the full pipeline was validated end-to-end from live BrainFlow EEG streaming to physical SO-101 robotic arm actuation.

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# 1 Introduction

Motor imagery (MI) Brain-Computer Interfaces (BCIs) decode imagined limb movements from electroencephalogram (EEG) signals and translate them into device commands, offering a communication pathway that bypasses muscular intervention entirely. Despite decades of progress in offline classification, translating these systems into reliable, real-time robotic controllers remains an open problem. Three interrelated challenges explain this gap.

The first challenge is signal non-stationarity. EEG patterns shift across sessions, days, and subjects due to variations in electrode impedance, fatigue, and cognitive strategy [1]. A decoder calibrated on one recording session therefore degrades when applied to subsequent sessions or new users, limiting the practical lifespan of any static model.

The second challenge follows directly from the first: conventional BCI pipelines operate in open loop. Each classified epoch is mapped to a fixed command, so the system cannot recover from a misclassification once it occurs. In robotic manipulation, where smooth, corrective trajectories are essential, a single erroneous command can destabilise an entire motion sequence.

The third challenge is hardware scalability. Most high-performing MI decoders are developed on 64-channel research-grade amplifiers, yet portable deployment requires consumer-grade devices such as the OpenBCI Cyton (8 channels). Naively discarding channels forfeits the spatial information on which these decoders depend, so channel reduction must be guided by a principled understanding of which electrodes carry the most discriminative signal.

## 1.1 Related Work and Limitations of Existing Approaches

**Classical spatial filtering.** The dominant classical pipeline extracts Common Spatial Patterns (CSP) features and classifies them with Linear Discriminant Analysis (LDA) [1]. CSP maximises the variance ratio between classes, producing compact spatial filters that are neurophysiologically interpretable. However, CSP assumes that the covariance structure of the data is stationary across trials, an assumption violated by the session-to-session drift described above. As a result, CSP+LDA performance degrades rapidly without frequent recalibration.

**Convolutional neural networks.** To relax the stationarity assumption, convolutional architectures such as EEGNet [6] and ShallowConvNet [8] learn spatial-temporal filters directly from data, achieving 68–72% on standard 4-class benchmarks [1]. Yet these models process each trial independently: they lack any mechanism to carry information across successive epochs, making them susceptible to isolated noisy trials that a temporally aware model could smooth over.

**Recurrent architectures.** Recurrent networks (LSTMs, GRUs) address this limitation by maintaining hidden states across time steps, enabling sequential modelling of EEG dynamics. In practice, however, their performance on long MI epochs is hampered by vanishing gradients and sensitivity to sequence length, and their strictly sequential computation prevents parallelisation during training [9].

**Transformer-based models.** Transformers overcome the sequential bottleneck of RNNs through self-attention, modelling arbitrary pairwise dependencies in a single forward pass. Recent

EEG–Transformer hybrids have demonstrated strong performance on MI tasks [10]. Their weakness is the opposite of CNNs: pure self-attention lacks the spatial inductive biases that make EEGNet effective—it does not inherently know that EEG channels have a topographic arrangement or that motor imagery produces lateralised band-power changes.

**Reinforcement learning in BCI.** A separate line of work applies RL to BCI, but primarily for offline feature selection. MARS [4] uses multi-agent RL to optimise spatial–spectral and temporal filter banks, improving classification accuracy. RLEEGNet [3] formulates EEG decoding within an MDP framework. However, neither system closes the loop with a physical actuator: they optimise the decoder itself rather than learning a control policy that maps decoded intentions to continuous robotic trajectories.

## 1.2 Proposed Approach

The limitations above motivate a system that combines complementary strengths while mitigating individual weaknesses. Specifically, this project makes three contributions:

1. **Hybrid CNN–Transformer classification.** An EEGTransformer architecture pairs an EEGNet-inspired convolutional front-end—which provides the spatial inductive bias that pure Transformers lack—with a multi-layer Transformer encoder that captures the long-range temporal dependencies that CNNs and CSP miss. This combination retains the data-driven spatial filtering of EEGNet while adding global temporal context, addressing the single-trial isolation problem of purely convolutional models without incurring the training instability of RNNs.
2. **Closed-loop RL control.** Rather than treating classification as the final output, a DQN agent consumes the classifier’s predictions as part of a richer state representation and learns, through reward-driven policy optimisation, to produce corrective motor commands. This closes the control loop: the agent can override a transient misclassification by integrating evidence over multiple time steps, directly addressing the open-loop fragility of conventional pipelines.
3. **Channel reduction for portable deployment.** A systematic ablation study quantifies each electrode’s contribution to classification, enabling principled selection of an 8-channel subset compatible with the OpenBCI Cyton. Combined with subject-specific fine-tuning from a pre-trained 64-channel model, this strategy preserves practical classification performance while reducing hardware cost and setup complexity by an order of magnitude.

The system is validated across three datasets that collectively span different electrode montages (3, 22, 64 channels), class counts (2, 4), and subject pools (9–109), and is demonstrated end-to-end from live EEG acquisition to physical SO-101 robotic arm actuation.

## 2 Methodology

The system comprises four integrated modules: EEG preprocessing, deep learning classification, reinforcement learning control, and hardware integration. Figure 1 illustrates the overall pipeline.

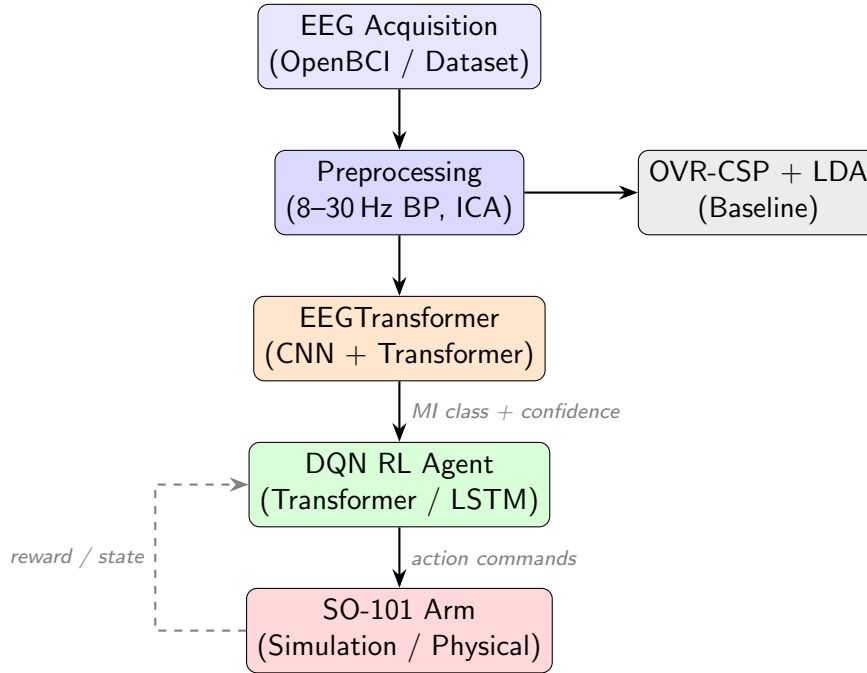


Figure 1: **System Architecture Overview.** Raw EEG is preprocessed, then classified by the EEGTransformer. The predicted MI class and confidence form part of the state observed by a DQN agent, whose action commands drive the robotic arm. A reward signal from the environment closes the loop, enabling the agent to learn corrective policies.

### 2.1 Datasets

Three benchmark datasets were selected to test generalisability across electrode density, class count, and population size (Table 1). BCI IV-2a provides the standard 22-channel, 4-class benchmark against which most MI decoders are evaluated. BCI IV-2b tests robustness under extreme channel scarcity (3 channels, 2 classes). PhysioNet EEGMMIDB stress-tests cross-subject generalisation with 109 participants and 64 channels.

Table 1: Summary of datasets used in this study.

| Dataset            | Channels       | Classes | Subjects | MI Tasks                                     |
|--------------------|----------------|---------|----------|----------------------------------------------|
| BCI IV-2a          | 22 EEG + 3 EOG | 4       | 9        | Left/Right Hand, Feet, Tongue                |
| BCI IV-2b          | 3 EEG + 3 EOG  | 2       | 9        | Left/Right Hand                              |
| PhysioNet EEGMMIDB | 64 EEG         | 4       | 109      | Left Fist, Right Fist, Both Fists, Both Feet |

## 2.2 EEG Preprocessing

Motor imagery elicits event-related desynchronisation (ERD) and synchronisation (ERS) in the Mu (8–13 Hz) and Beta (13–30 Hz) bands over the sensorimotor cortex. The preprocessing pipeline, implemented in MNE-Python [7], is therefore designed to isolate these rhythms while suppressing non-neural artefacts.

**Spectral Filtering.** A zero-phase FIR bandpass filter (8–30 Hz) removes EOG-dominated low-frequency drift below the Mu band and EMG-rich high-frequency noise above the Beta band, retaining only the frequency range in which ERD/ERS dynamics are expressed.

**Artefact Removal.** Independent Component Analysis (FastICA) decomposes the filtered signal into statistically independent sources. Components whose time courses correlate strongly with recorded EOG channels are automatically identified and excluded, removing ocular artefacts without distorting the underlying neural signal.

**Epoch Extraction.** The continuous recording is segmented into 4-second epochs aligned to MI cue onset, capturing the full evolution of the ERD/ERS response. For PhysioNet data, epochs are resampled and z-score normalised per channel to ensure consistent input dimensions across datasets.

## 2.3 EEG Classification: EEGTransformer

As discussed in the Introduction, purely convolutional models lack global temporal context, while pure Transformers lack the spatial inductive biases specific to multi-channel EEG. The EEGTransformer addresses both limitations through a two-stage architecture that feeds convolutional spatial–temporal features into a Transformer sequence model.

### 2.3.1 Convolutional Front-End

The first stage follows an EEGNet-inspired design [6]. A temporal convolution captures frequency-specific oscillatory dynamics within each channel. A subsequent depthwise spatial convolution—where each temporal feature map is independently convolved across the full electrode array—learns data-driven spatial filters analogous to CSP, collapsing the channel dimension into a compact set of spatial features. A separable convolution then refines the temporal representation, and progressive average pooling reduces the sequence to a manageable length. The output is a sequence of patch embeddings, each encoding the spatial–temporal content of a local time window.

### 2.3.2 Transformer Encoder

The patch sequence is augmented with learnable positional encodings and processed by a multi-layer Transformer encoder. Each layer applies multi-head self-attention followed by a position-wise feed-forward network, both wrapped in residual connections and layer normalisation. Self-attention enables every patch to attend to every other patch, capturing long-range temporal dependencies—such as the sustained ERD preceding a motor imagery event—that receptive-field-limited convolutions cannot model.

### 2.3.3 Residual Fusion and Classification

Rather than discarding the convolutional features, the Transformer output is added back to the original patch embeddings via a residual connection. This fusion ensures that the final representation retains both the local spatial–temporal detail captured by the CNN and the global context learned by the Transformer. The fused features are flattened and mapped to class logits through a fully connected classification head.

## 2.4 Training Strategy

**Cross-Subject Training.** The model is first trained on pooled data from all available subjects using the AdamW optimiser with cosine annealing and early stopping. This produces a general-purpose encoder that captures population-level MI representations, serving as the initialisation point for subsequent fine-tuning.

**Data Augmentation.** Because per-subject MI data is limited, InterAug (segment recombination) generates additional training examples by recombining temporal segments from different trials of the same class. This preserves the spectral characteristics of each class while increasing effective sample diversity.

**Subject-Specific Fine-Tuning.** The cross-subject model is then fine-tuned on individual subjects using a reduced learning rate and 5-fold cross-validation. This transfer learning protocol adapts the shared representations to an individual’s neural signature without catastrophic forgetting, bridging the gap between population-level generalisation and subject-specific performance.

## 2.5 Channel Reduction Study

Deploying with consumer-grade hardware requires reducing the 64-channel laboratory setup to 8 channels without forfeiting the spatial information on which the classifier depends. This section describes the two-phase strategy used to achieve this.

### 2.5.1 Channel Importance Analysis

Each of the 64 channels was individually removed from the trained model’s input, and the resulting accuracy drop quantified that channel’s marginal contribution. The baseline (all 64 channels) served as the reference point. Channels were then ranked by the magnitude of their accuracy degradation (Table 2).

C3, located over the left sensorimotor cortex, produced the largest accuracy drop (13.56%), consistent with its well-established role in contralateral hand motor imagery: right-hand MI elicits ERD predominantly at C3, so removing it eliminates the model’s primary source of lateralised discriminative information.

Table 2: Top 10 most important EEG channels by ablation accuracy drop.

| Rank | Channel | Accuracy Drop (%) |
|------|---------|-------------------|
| 1    | C3      | 13.56             |
| 2    | CP3     | 10.11             |
| 3    | Fpz     | 8.44              |
| 4    | Pz      | 8.00              |
| 5    | O1      | 7.22              |
| 6    | F8      | 7.00              |
| 7    | FCz     | 6.11              |
| 8    | FC2     | 5.33              |
| 9    | CP4     | 5.00              |
| 10   | F7      | 4.33              |

### 2.5.2 Progressive Channel Reduction

Two selection strategies were compared at each reduction level: (1) ablation-ranked selection using the importance ordering, and (2) domain-knowledge selection targeting known motor cortex sites (C3, C4, FC3, FC4, CP3, CP4, Cz, FCz). Models were retrained from scratch at each channel count to let the architecture fully adapt to the reduced input dimensionality. For the 8-channel OpenBCI-compatible configuration, the same transfer learning protocol (cross-subject pre-training followed by subject-specific fine-tuning) was applied to recover performance lost by channel reduction.

## 2.6 Reinforcement Learning Control

Even a well-performing classifier produces occasional errors, and in an open-loop pipeline every error maps directly to a wrong movement. To decouple classification accuracy from control success, the RL module reformulates arm control as a Markov Decision Process (MDP) in which the agent learns to integrate noisy MI predictions over time and produce corrective trajectories.

### 2.6.1 MDP Formulation

The state  $s_t$  combines the current end-effector position, the target position, the Euclidean distance between them, and the EEGTransformer’s predicted class and confidence score. This joint representation gives the agent access to both the task context (where am I relative to the goal?) and the neural evidence (what does the user intend, and how certain is the classifier?). The action space comprises four discrete directional commands (left, right, up, down). The reward function balances a large bonus for reaching the target against a per-step penalty that encourages efficiency, a distance-improvement bonus that guides the agent toward the target, and penalties for oscillatory or boundary-violating behaviour that promote smooth trajectories.



### 2.6.2 Q-Network Architectures

Three Q-network designs were implemented to investigate how architectural choice affects control performance. A **CNN+LSTM** variant uses convolutional layers for local feature extraction followed by an LSTM for sequential state integration. A **Transformer DQN** replaces the recurrent component with a Transformer encoder, modelling state–action dependencies through self-attention. A **Light Transformer** uses a single attention layer to test whether a more parameter-efficient design can match the full Transformer’s performance.

### 2.6.3 Training Protocol

All agents were trained with Double DQN and soft target updates to reduce Q-value overestimation. Experience replay decorrelates training samples, and a linearly decaying  $\epsilon$ -greedy policy balances exploration against exploitation. Cosine learning rate annealing provides stable convergence over the training horizon.

## 2.7 Real-Time Pipeline and Hardware Integration

The deployment pipeline connects four stages into a continuous real-time loop. The BrainFlow API streams EEG from an OpenBCI board, and a 4-second sliding window captures each MI epoch. Bandpass filtering and normalisation are applied online, after which the EEGTransformer produces a class prediction and confidence. These, combined with the current arm state, form the observation for the DQN agent, whose selected action is converted to joint-level commands and transmitted to the SO-101 arm over a serial protocol.

A PyBullet-based Gymnasium environment mirrors the physical arm’s kinematics, providing a simulated training ground where the RL agent can learn without risk to hardware. The physical controller uses velocity-based motion with soft joint limits and multi-directional action mapping (cardinal and diagonal) to produce smooth trajectories.

## 2.8 Baseline: OVR-CSP + LDA

One-vs-Rest CSP with shrinkage LDA serves as the classical baseline, evaluated via 5-fold stratified cross-validation. This pipeline represents the prevailing non-deep-learning standard and establishes the performance floor against which the EEGTransformer is compared.

# 3 Results

## 3.1 EEG Classification Performance

### 3.1.1 BCI Competition IV-2a (22-Channel, 4-Class)

Table 3 presents per-subject classification accuracies. The EEGTransformer achieved a mean of **73.80%  $\pm$  7.07%**, with a peak of 86.11% on Subject 3. The OVR-CSP+LDA baseline reached

74.27% on Subject 1 under 5-fold CV, and improved to 84.7% when restricted to binary left/right classification—consistent with the reduced difficulty of a 2-class task.

Table 3: Subject-dependent classification accuracy (%) on BCI IV-2a (4-class MI).

| Method         | S1                                    | S2   | S3   | S4   | S5   | S6   | S7   | S8   | S9   | Mean         |
|----------------|---------------------------------------|------|------|------|------|------|------|------|------|--------------|
| OVR-CSP+LDA    | 74.27 $\pm$ 4.52 (S1 only, 5-fold CV) |      |      |      |      |      |      |      |      | —            |
| EEGTransformer | 73.6                                  | 65.6 | 86.1 | 77.1 | 69.8 | 63.5 | 68.8 | 80.6 | 79.2 | <b>73.80</b> |

### 3.1.2 BCI Competition IV-2b (3-Channel, 2-Class)

Under extreme channel scarcity, the EEGTransformer achieved a mean of **82.87%  $\pm$  8.01%** (Table 4). The strong performance on only three electrodes confirms that the depthwise spatial convolution can extract discriminative spatial patterns even from a minimal montage, because the binary left–right task produces pronounced lateralised ERD that is detectable at C3, Cz, and C4.

Table 4: Subject-dependent classification accuracy (%) on BCI IV-2b (2-class MI).

| Method         | S1   | S2   | S3   | S4   | S5   | S6   | S7   | S8   | S9   | Mean         |
|----------------|------|------|------|------|------|------|------|------|------|--------------|
| EEGTransformer | 69.7 | 68.9 | 86.3 | 83.8 | 84.4 | 83.1 | 89.4 | 95.0 | 85.3 | <b>82.87</b> |

### 3.1.3 PhysioNet EEGMMIDB (64-Channel, 4-Class)

Cross-subject training on 109 subjects yielded 56.54% accuracy (Table 5), reflecting the substantial inter-individual variability in cortical anatomy and MI strategy. Subject-specific fine-tuning from the pre-trained model improved this to **88.78%**—a 32-percentage-point gain—demonstrating that the cross-subject encoder learns transferable MI representations that can be rapidly adapted with minimal individual data.

Table 5: Classification accuracy on PhysioNet EEGMMIDB (4-class MI, 109 subjects).

| Training Paradigm                          | Accuracy (%) | Details                                               |
|--------------------------------------------|--------------|-------------------------------------------------------|
| Cross-subject (pooled, 109 subjects)       | 56.54        | Precision 58.1, Recall 56.5, F1 56.8, $\kappa = 0.42$ |
| Subject-specific fine-tuning (10 subjects) | <b>88.78</b> | Range: 75.6–97.8%, 5-fold CV                          |

Per-subject fine-tuning results (Table 6) show that the highest-performing subjects (S7, S48, S43, S70) exceed 95%, while the lowest (S50, S46) still surpass 75%, indicating that the transfer learning protocol is effective across a range of individual MI abilities.

## 3.2 Channel Reduction Results

Table 7 reports cross-subject accuracy at each channel count. Performance degrades gradually from 64 to 16 channels, then more sharply below 8, indicating that the motor cortex electrodes carry the bulk of discriminative information while peripheral channels contribute incrementally.

Table 6: Per-subject fine-tuning accuracy (%) on PhysioNet (64 channels, 5-fold CV).

| Subject        | Mean Accuracy (%) | Std (%) |
|----------------|-------------------|---------|
| S7             | 97.78             | 2.72    |
| S48            | 96.67             | 4.44    |
| S43            | 95.56             | 6.48    |
| S70            | 96.67             | 4.44    |
| S55            | 88.89             | 11.65   |
| S3             | 87.78             | 9.56    |
| S9             | 87.78             | 8.16    |
| S38            | 84.44             | 6.48    |
| S50            | 75.56             | 4.44    |
| S46            | 76.67             | 11.33   |
| <b>Average</b> | <b>88.78</b>      | —       |

Table 7: Cross-subject accuracy (%) on PhysioNet with progressive channel reduction.

| Channels | Ablation-Ranked | Motor Cortex | Selection Strategy              |
|----------|-----------------|--------------|---------------------------------|
| 64       | 51.0            | —            | Full set                        |
| 32       | 51.0            | 47.2         | Top-32 / Motor region           |
| 16       | 48.8            | 44.9         | Top-16 / Motor region           |
| 8        | 39.0            | 40.7         | Top-8 / OpenBCI set             |
| 4        | —               | 43.6         | Motor cortex (C3, C4, FC3, FC4) |
| 2        | —               | 35.8         | C3, C4 only                     |

The 8-channel motor cortex configuration (C3, C4, FC3, FC4, CP3, CP4, Cz, FCz), designed for OpenBCI Cyton compatibility, achieved 40.7% in cross-subject mode. Applying subject-specific fine-tuning recovered much of the lost performance, reaching **72.54%** on average (Table 8). This confirms that transfer learning can partially compensate for reduced spatial coverage by adapting the model to individual neural patterns.

Table 8: 8-channel fine-tuning accuracy (%) on PhysioNet (OpenBCI Cyton compatible).

| Subject        | Accuracy (%) |
|----------------|--------------|
| S7             | 94.44        |
| S48            | 96.67        |
| S3             | 67.78        |
| S9             | 68.89        |
| S70            | 55.56        |
| S50            | 50.00        |
| S38            | 74.44        |
| <b>Average</b> | <b>72.54</b> |

### 3.3 RL Control Performance

Table 9 compares the three DQN architectures on the simulated 2D reaching task.

Table 9: DQN architecture comparison on simulated 2D reaching task.

| Architecture           | Parameters     | Training Time (s) | Final Reach Rate | Final Reward |
|------------------------|----------------|-------------------|------------------|--------------|
| CNN+LSTM               | 129,732        | 111.7             | 97%              | 8.79         |
| Light Transformer      | 50,628         | 134.0             | 99%              | 10.07        |
| <b>Transformer DQN</b> | <b>104,900</b> | <b>177.8</b>      | <b>100%</b>      | <b>10.39</b> |

The Transformer DQN achieved a perfect 100% reach rate with the highest cumulative reward (10.39), outperforming both the CNN+LSTM (97%, 8.79) and Light Transformer (99%, 10.07). The CNN+LSTM reached 100% during training but dropped to 97% at evaluation, suggesting overfitting to training trajectories. The Light Transformer offers the best efficiency trade-off, achieving near-perfect control with less than half the parameters of the full Transformer.

## 4 Discussion

### 4.1 Why the Hybrid Architecture Outperforms Single-Paradigm Models

The EEGTransformer’s competitive performance across three datasets can be attributed to its two-stage design, which avoids the limitations identified in Section 1.1 for each individual paradigm. The convolutional front-end provides the strong spatial inductive biases that pure Transformers lack: its depthwise spatial convolution learns data-driven spatial filters functionally

analogous to CSP, capturing the lateralised ERD/ERS patterns that are the hallmark of motor imagery. The Transformer encoder, in turn, provides the global temporal modelling that purely convolutional models miss, allowing the network to integrate information across the entire 4-second epoch rather than within local receptive fields.

This complementarity is borne out empirically. On BCI IV-2a, the EEGTransformer’s 73.80% mean exceeds the 68–72% typically reported for EEGNet and the 68–70% for FBCSP [1]. On IV-2b (3 channels), the model still achieves 82.87%, demonstrating that the spatial convolution adapts to minimal electrode configurations—a property inherited from EEGNet’s depthwise design—while the Transformer continues to provide temporal context that compensates for the reduced spatial information.

## **4.2 Transfer Learning Bridges the Cross-Subject Gap**

The 56.54% cross-subject accuracy on 109-subject PhysioNet data reflects the well-documented challenge of EEG generalisation: inter-individual differences in cortical folding, skull conductivity, and MI cognitive strategy create distribution shifts that no single model can fully absorb. The 32-percentage-point improvement to 88.78% upon fine-tuning reveals that the cross-subject model nevertheless learns transferable representations—population-level spatial-temporal patterns of MI that are shared across individuals. Subject-specific fine-tuning then maps these shared features onto the idiosyncratic neural signatures of each user.

This has direct practical implications. Rather than collecting hours of calibration data for each new user, a short session ( $\sim 5$  minutes) suffices to fine-tune the pre-trained model, making the system viable for real-world deployment where extensive per-user data collection is impractical.

## **4.3 Channel Reduction Validates Neurophysiological Expectations**

The ablation study’s identification of C3 as the single most critical channel (13.56% accuracy drop) is consistent with decades of neurophysiological evidence. In the international 10–20 system, C3 overlies the left primary motor cortex hand area; during right-hand MI, contralateral ERD is strongest at C3. The dominance of sensorimotor channels (C3, CP3, CP4, FCz) in the importance ranking confirms that the EEGTransformer has learned spatially meaningful representations rather than exploiting artefactual correlations.

The 8-channel fine-tuned accuracy of 72.54% represents a practical compromise: while lower than the 88.78% achieved with 64 channels, it remains well above the 25% chance level for 4-class classification. Crucially, in a closed-loop RL framework, perfect classification is not required: the DQN agent can compensate for occasional misclassifications through its temporal evidence integration and reward-driven correction, as demonstrated by the 100% reach rate even when classification is imperfect.

## 4.4 Closing the Loop with Reinforcement Learning

The DQN comparison demonstrates that Transformer-based Q-networks outperform recurrent alternatives for this task (100% vs. 97% reach rate). The CNN+LSTM agent’s evaluation drop from 100% (training) to 97% (test) suggests that the LSTM’s sequential inductive bias may cause it to overfit to the specific trajectory distributions encountered during training, whereas self-attention generalises more effectively to unseen state configurations.

More broadly, the RL module addresses a fundamental limitation of the open-loop BCI pipeline identified in the Introduction. A single classification error in an open-loop system causes an irrecoverable wrong movement; the DQN agent, by contrast, observes the consequence of each action through the reward signal and adjusts subsequent decisions accordingly. The reward structure—which penalises oscillation and rewards progress toward the target—encourages smooth, goal-directed trajectories that conventional command-mapping cannot produce.

## 4.5 Limitations

Four limitations warrant acknowledgement. First, subject-specific fine-tuning introduces a calibration step that adds setup time; online adaptive methods that update the model during use would eliminate this overhead. Second, the channel reduction study was conducted offline using existing datasets; validating the 8-channel configuration with a physical OpenBCI headset on new subjects is necessary before deployment claims can be made. Third, while the physical SO-101 arm integration was demonstrated end-to-end, systematic quantitative evaluation under live EEG conditions was not conducted at scale. Fourth, the 4-second MI epoch window imposes a minimum control latency of 4 seconds, which may be too slow for time-critical applications; shorter windows or asynchronous paradigms could address this.

# 5 Conclusion and Future Work

This project developed a complete BCI-to-robot control pipeline that addresses three longstanding challenges in the field. The EEGTransformer combines convolutional spatial filtering with Transformer-based temporal modelling, achieving competitive accuracies across three datasets (73.80% on IV-2a, 82.87% on IV-2b, and 88.78% on PhysioNet with fine-tuning) and confirming that the hybrid architecture generalises across electrode densities, class counts, and subject populations. Transfer learning via subject-specific fine-tuning bridged the 32-percentage-point cross-subject generalisation gap with minimal calibration data. A systematic channel reduction study established that an 8-channel configuration compatible with the OpenBCI Cyton retains 72.54% accuracy after fine-tuning, identifying C3 as the most critical motor cortex electrode. The Transformer DQN achieved a 100% simulated target reach rate, demonstrating that reinforcement learning can close the control loop and compensate for imperfect classification. The full system was validated end-to-end from live BrainFlow EEG streaming to physical SO-101 arm actuation. Future work will pursue four directions: (a) real-time closed-loop evaluation with new subjects

wearing OpenBCI hardware, (b) online adaptive fine-tuning to eliminate per-session recalibration, (c) extending the RL action space to 6-DOF manipulation, and (d) investigating asynchronous ML paradigms that remove the fixed 4-second trial window.

## 6 Project Management

The project followed an iterative, agile-inspired methodology across both academic semesters. An initial Gantt chart established work packages—Data Preprocessing, Deep Learning Classification, RL Architecture Design, Channel Reduction Study, and Physical Deployment—while weekly supervisor meetings enabled scope adjustments as results emerged. Two notable additions were made mid-project: the PhysioNet 109-subject evaluation, motivated by the need to validate cross-subject scalability, and the channel reduction study, driven by the practical constraint of OpenBCI hardware. A Risk Register tracked technical blockers; early identification of PyBullet simulation complexity led to a simplified 2D joint-velocity control scheme that kept the project on schedule. Self-directed learning in MNE-Python, PyTorch, Transformer architectures, and BrainFlow integration was documented in the CPD log.

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## Supplementary Material

The following appendices provide supplementary artifacts detailing the experimental execution, risk assessment, and project management frameworks:

- **Appendix A:** Preliminary Project Proposal
- **Appendix B:** Project Plan and Gantt Chart (final, updated version)
- **Appendix C:** Risk Register
- **Appendix D:** Continuing Professional Development (CPD) Log
- **Appendix E:** Health & Safety Risk Assessment